

Project Design Phase

Problem – Solution Fit Template

Date	15 February 2025
Team ID	HYD-METRO-2025
Project Name	Hyderabad Metro Rail – Digital Ticketing System (ServiceNow)
Maximum Marks	2 Marks

About the Problem–Solution Fit

The Problem-Solution Fit means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why. The canvas below is fully filled for the Hyderabad Metro Rail Digital Ticketing project.

Problem–Solution Fit Canvas

1. CUSTOMER SEGMENT(S) C S	6. CUSTOMER CONSTRAINTS C C	5. AVAILABLE SOLUTIONS A S
• Daily metro commuters in Hyderabad (students, working professionals aged 18–50) • Metro station managers & operations staff • IT administrators managing the ServiceNow platform • Metro card holders seeking digital recharge	• Limited awareness that digital metro booking exists • Dependency on physical token counters (no alternative channel) • Cash-only payment — no UPI/card/wallet integration • No smartphone app available for metro ticketing • Connectivity issues in underground metro stations • Non-tech-savvy commuters unfamiliar with service portals	• Physical token counters (manual, cash-only, queue-dependent) • Smart card (partial — only for pre-purchased cards, no online recharge) • Third-party apps (unreliable, not officially endorsed by HMRL) • No official mobile app or web booking portal currently exists • Workaround: commuters arrive early to account for queue time

2. JOBS-TO-BE-DONE / PROBLEMS	J & P;	9. PROBLEM ROOT CAUSE	R C	7. BEHAVIOUR	B E
• Buy a metro ticket quickly without standing in a queue • Know the fare upfront before committing to the journey • Pay using UPI, card, or wallet — not cash • Get a digital ticket/QR code usable at the metro gate • Recharge metro card without visiting a counter • View journey history and booking records • Book tickets in advance from home or office		• HMRL has not implemented an Automated Fare Collection (AFC) system • No digital ticketing platform or mobile app has been built • All booking is counter-dependent, manual, and cash-based • No backend database or API to support digital integration • Legacy operations not yet digitized under Digital India initiative • Budget and priority constraints delayed technology adoption		• Arrives 10–15 mins early every day to account for queues • Carries exact cash to avoid change problems at counters • Switches to Ola/Uber/auto when queue time is too long • Checks Google Maps for train times but still queues for ticket • Uses UPI/GPay everywhere else — frustrated it doesn't work here • Complains on social media about lack of digital ticketing	

3. TRIGGERS	10. YOUR SOLUTION	8. CHANNELS OF BEHAVIOUR
• Missing a train while standing in queue • Seeing friends use Bangalore/Delhi Metro apps • Digital India campaign awareness on social media • Fare confusion/error at counter causes delay • Station staff announces digital option available • Receiving a WhatsApp forward about metro app	• ServiceNow-Powered Digital Metro Ticketing • Service Portal (/sp) with 'Book A Metro Ticket' catalog item • Auto fare calculation via Catalog Client Script • QR code generated on submission via GlideModal • Multi-mode payment: UPI, Card, Wallet, Cash, Others • Flow Designer stores all bookings in Metro Database • Role-based ACL protects booking data • Email notification confirms every booking • Metro card recharge option in same portal	8.1 ONLINE • ServiceNow Service Portal (/sp) on browser • Email (booking confirmation, notifications) • Social media (Twitter/X, WhatsApp for discovery) • Google Maps (checking metro timings) • UPI apps (GPay, PhonePe — desired integration) 8.2 OFFLINE • Metro station kiosks and digital signage • Token counter staff (current channel) • QR scanner at metro gate turnstile • Printed flyers about digital booking
4. EMOTIONS: BEFORE / AFTER BEFORE (Problem): • Stressed & frustrated every morning by queues • Anxious about missing the train • Embarrassed without exact change • Helpless — no alternative exists AFTER (Solution): • Relieved — ticket booked in < 60 seconds • Confident — QR code in hand before leaving home • Empowered — UPI payment accepted • Delighted — zero queue at gate		

Problem–Solution Fit Validation Summary

Fit Dimension	Problem	ServiceNow Solution	Fit?
Customer Segment	Hyderabad Metro daily commuters	Service Portal accessible to any authenticated user	✓ Strong Fit
Job-to-be-Done	Book ticket quickly without queue	Catalog form bookable in < 60 seconds from any device	✓ Strong Fit
Constraint: Cash-only	No digital payment channel	Payment mode variable (UPI/Card/Wallet/Cash/Others)	✓ Strong Fit
Constraint: No app	No official booking channel exists	ServiceNow Service Portal — no app install needed (browser)	✓ Strong Fit
Behaviour: Uses UPI daily	Can't use UPI for metro	Payment mode captures UPI intent; gateway planned for 2020	✓ Strong Fit
Trigger: Missed train	Queue caused delay	Pre-booking eliminates queue — book before leaving home	✓ Strong Fit
Emotion: Frustrated → Delighted	Signified of daily queue	QR code ready before commute; gate scan in < 2 seconds	✓ Strong Fit

Channel: Online (browser)	No online channel existed	ServiceNow /sp portal — browser-based, mobile-responsive	Positive Fit
Root Cause: No AFC system	Manual only ticketing	ServiceNow automates end-to-end: form→fare→QR→DB→orig fail	Strong fit